Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 12)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows some events in the history of life on Earth.

Table 1

Event	Time before present / billion years
Formation of the Earth	4.6
First life (Archaea) on Earth	3.5
First photosynthetic organisms	2.7
Oxygen concentration in the atmosphere starts to rise	2.45
End of the period of oxygen build-up (Proterozoic)	0.54

0 1 . 1 Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

Answer _____ million years

0 1 . 2 Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

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0 1 . 3 Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

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0 1 . 4 Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

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0 1 . 5 Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

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0 2 . 1 Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

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0 2 . 2 Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

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0 2 . 3 Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

Blue whale .....

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Large blue butterfly .....

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0 3 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 3 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 4 Table 1 shows features of some fishing methods.

0 4 . 1 Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage	
Demersal trawling			
Purse seining			
Pelagic long-lining			

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0 4 . 2 Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

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0 4 . 3 Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

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0 5 . 1 Describe **four** ways in which the Wildlife and Countryside Act (1981) protects wildlife in the UK.

[4 marks]

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0 5 . 2 Explain the difference between CITES Appendix I and Appendix II, giving an example species for each.

[4 marks]

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0 5 . 3 Explain why controlling international trade can be more effective than trying to stop collection of a species in its own country.

[2 marks]

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0 6 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 6 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 6 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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0 7 . 1 Explain **three** features of natural systems that make them sustainable.

[3 marks]

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0 7 . 2 Explain why the human use of mineral resources makes their sustainable exploitation much harder than the use of materials in a natural cycle.

[2 marks]

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0 7 . 3 Explain **three** principles of the circular economy that should be applied when a new product is designed.

[3 marks]

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0 7 . 4 Explain why an activity that is sustainable for a small population may be unsustainable for a large one. Use **one** example.

[3 marks]

0 7 . 5 Explain why urban populations are less likely than rural communities to notice that a resource is being over-exploited, and give **two** consequences of this.

[4 marks]

15

0 8 . 1 Compare selective breeding, asexual reproduction and cross-breeding as methods of improving livestock or crops.

[6 marks]

0 8 . 2 Explain how artificial insemination and embryo transfer increase the rate at which desirable characteristics spread through a livestock population.

[4 marks]

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0 9 . 1 Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

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0 9 . 2 Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

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0 9 . 3 Explain how auditory monitoring can be used to study bats.

[2 marks]

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0 9 . 4 Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

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10.1 Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **biodiversity**.

[1 mark]

10

Write an essay on one of the following topics.

11 . 1 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

OR

11 . 2 Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS